Multi-Year Tax Simulation Report

1. Introduction

The **Multi-Year Tax Simulation** program is designed to calculate and analyze income tax over multiple years. It simulates how tax liability changes when income increases annually by a fixed growth rate. The program also categorizes each year based on tax burden and evaluates the overall taxpayer status at the end of the simulation.

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2. Objective

The main objectives of the program are:

- To calculate yearly tax based on progressive tax slabs
- To simulate income growth over a specified number of years
- To compute total tax paid across all years
- To classify tax burden annually and overall

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3. Input Requirements

The program requires two user inputs:

1. **Number of years** – Total duration for tax simulation
2. **Initial income** – Starting annual income

Both inputs are converted into integers for calculation purposes.

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4. Tax Structure Used

The tax is calculated using the following slab system:

Income Range	Tax Rate
Up to 250,000	0%
250,001 – 500,000	5%
500,001 – 1,000,000	20%
Above 1,000,000	30%

The tax is calculated on the full income based on the slab in which it falls.

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5. Program Logic

The program uses a `while` loop to simulate each year. For every iteration:

1. The tax is calculated according to the income slab.
2. The yearly tax is added to the cumulative total tax.
3. The income is increased by 10% for the next year.
4. The year is categorized:
    - If tax > 100,000 → **High tax year**
    - Otherwise → **Normal tax year**

After completing all years:

- If total tax > 500,000 → **Heavy taxpayer**
- Otherwise → **Moderate taxpayer**

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6. Output

For each year, the program displays:

- Year number
- Tax calculated
- Tax category (High or Normal)

At the end, it displays:

- Total tax paid
- Overall taxpayer classification

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7. Key Features

- Uses looping for multi-year simulation
- Implements conditional statements for slab-based taxation
- Applies annual income growth automatically
- Tracks cumulative tax efficiently

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8. Error Corrections Made

During debugging, the following issues were fixed:

- Corrected indentation inside the loop
- Added missing colon in an `if` statement
- Closed missing parenthesis in a print statement

- Ensured proper loop structure

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9. Conclusion

The Multi-Year Tax Simulation program effectively demonstrates the use of loops, conditional statements, arithmetic operations, and cumulative tracking in Python. It provides a practical example of how income growth impacts taxation over time and helps categorize taxpayers based on financial burden.

This program is useful for understanding progressive tax systems and long-term financial projections.